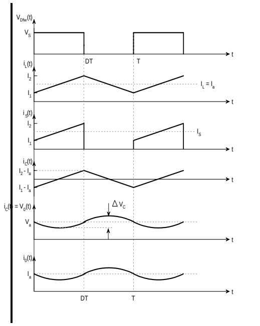
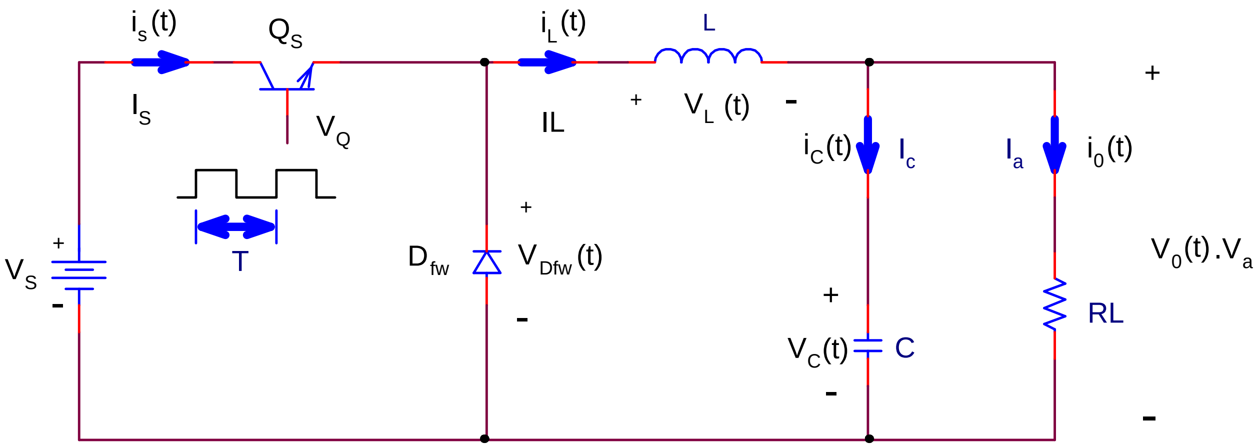
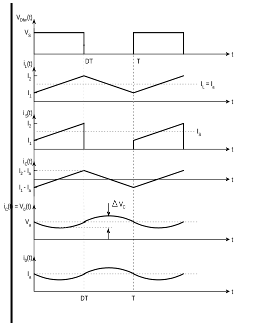
Buck converter



- The output voltage is smaller than the input voltage
- Same polarity



Buck converter
Continuous mode

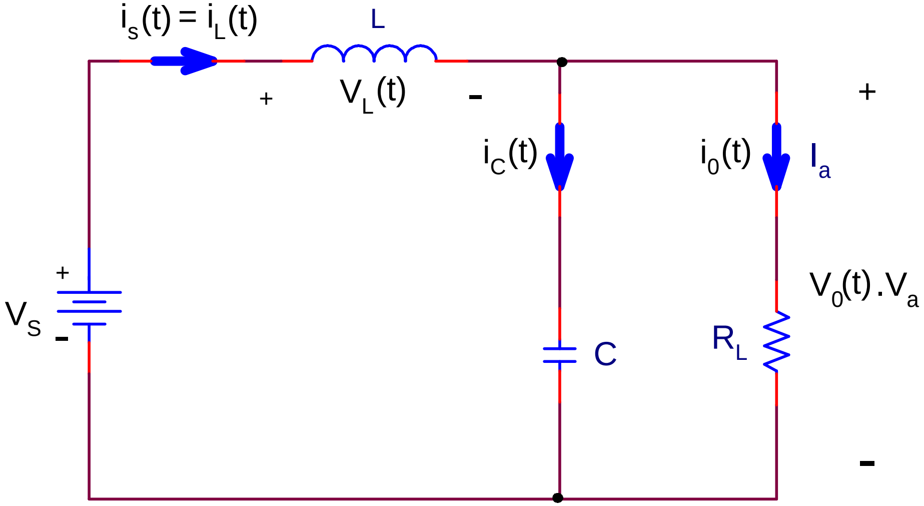


- *MODE 1 (0 < t < t_on)*

$$v_L(t) = L\frac{di}{dt} \;,$$

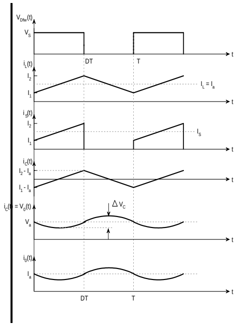
$$V_s - V_a = L\;\frac{I_2 - I_1}{t_{on}} = L\;\frac{\Delta I}{t_{on}}\;,$$

$$t_{on} = \frac{L\;\;\Delta I}{(V_s - V_a)}\;.$$



Buck converter

Continuous mode



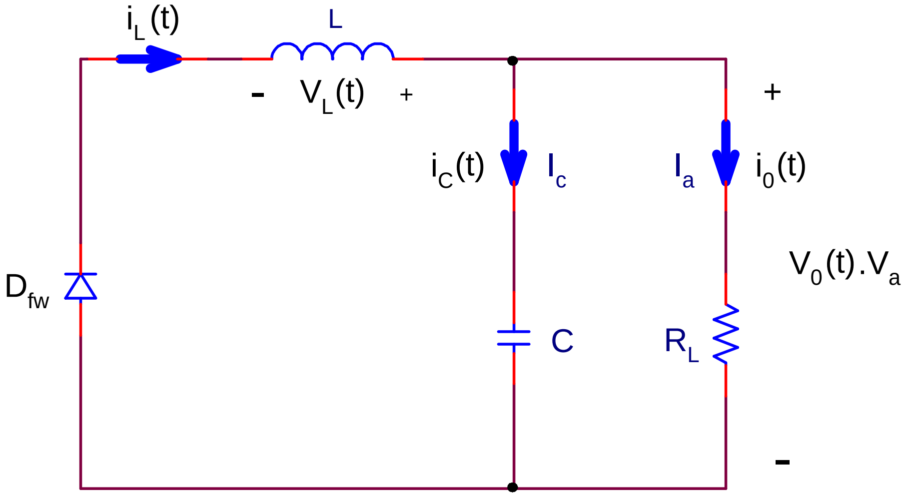
- *MODE 2 (t_{on} < t < T)*

$$-V_a = L \; \frac{I_1 - I_2}{t_{off}} \;,$$

$$V_a = L \; \frac{\Delta I}{t_{off}} \;.$$

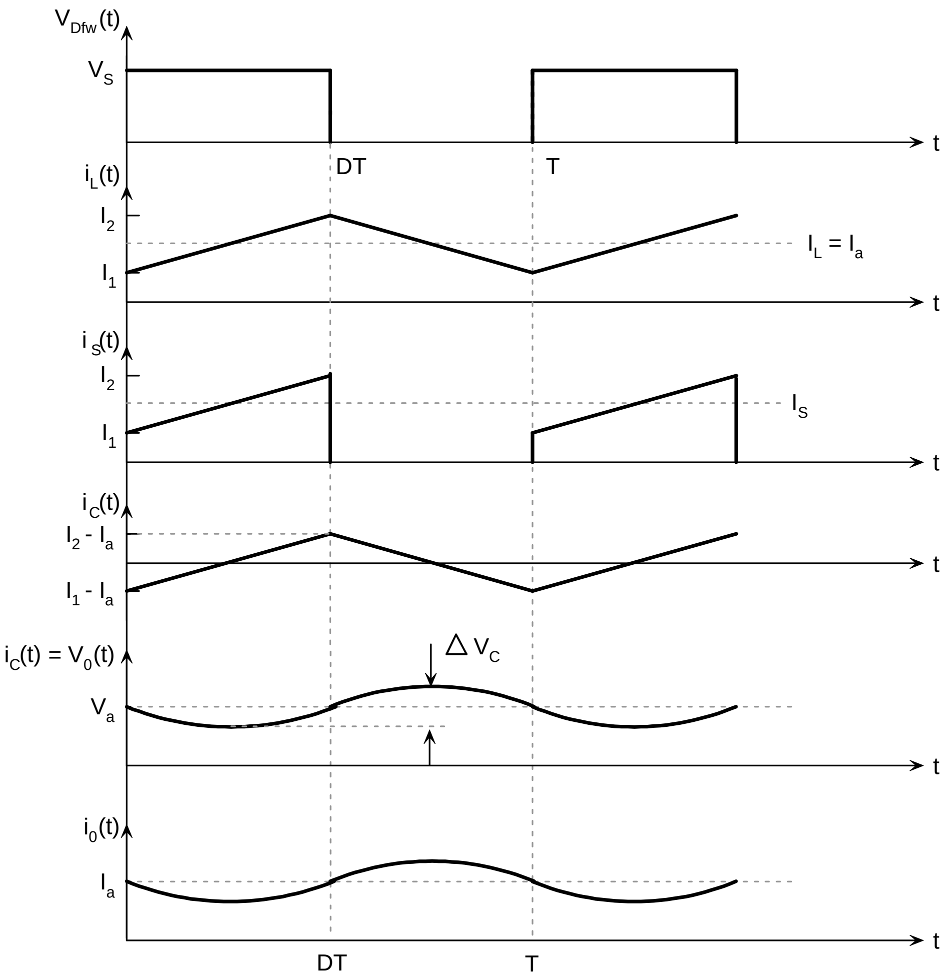
$$t_{off} = \frac{L \Delta I}{V_a} \;,$$

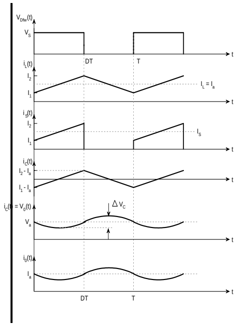
$$\Delta I = \frac{V_a \;\; t_{off}}{L} \;.$$

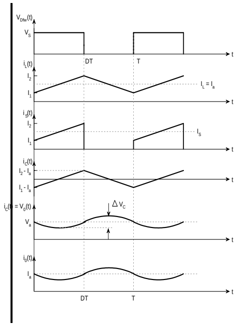


Buck converter

Continuous mode







Buck converter

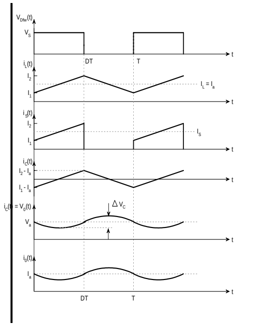
Continuous mode

- Voltage conversion ratio

$$\Delta I = \frac{(V_s - V_a)\ t_{on}}{L} = \frac{V_a\ t_{off}}{L}\ .$$

$$(V_s - V_a)\ DT = V_a\ (1 - D)\ T$$

$$V_a = \frac{V_s\ DT}{T} = V_s\ D\ .$$



Buck converter

Continuous mode

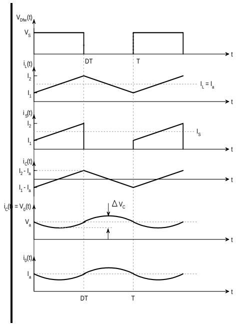
- Inductor ripple current

$$T = \frac{1}{f_s} = t_{on} + t_{off}$$

$$= \frac{L\,\Delta I}{V_s - V_a} + \frac{L\,\Delta I}{V_a} = \frac{L V_s\,\Delta I}{V_a\left(V_s - V\right)_a}$$

$$\Delta I = \frac{V_a\,(V_s - V_a)T}{L V_s} = \frac{D V_s\,(1 - D)}{f_s\,L} \;.$$

Buck converter
Continuous mode



- Capacitor current
  - The average capacitor current I_c is zero

$$i_L = i_c + i_0$$

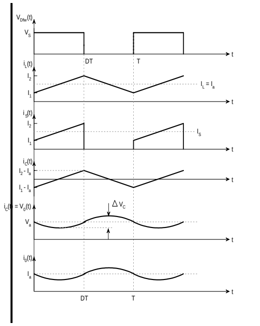
$$\Delta i_L = \Delta i_c + \Delta i_o \approx \Delta i_c$$

$$\frac{t_{on}}{2} + \frac{t_{off}}{2} = \frac{T}{2}$$

$$I_c = \frac{\Delta I}{4} \; .$$

Buck converter

Continuous mode

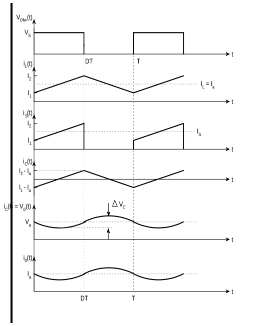


- Capacitor voltage
  - The average capacitor voltage is V_a

$$v_c(t) = v_c(0) + \frac{1}{C}\int_0^{\frac{T}{2}} i_c(t)\,dt$$

$$\Delta v_c = v_c(t) - v_c(0) = \frac{1}{c}\int_0^{\frac{T}{2}} \frac{\Delta I}{4}\,dt = \frac{T\,\Delta I}{8C} = \frac{\Delta I}{8 f_s C}$$

$$\Delta v_c = \frac{DV_s(1-D)}{f_s L}\;\frac{1}{8 f_s C} = \frac{V_s D(1-D)}{8 f_s^2 LC}\;.$$



Buck converter

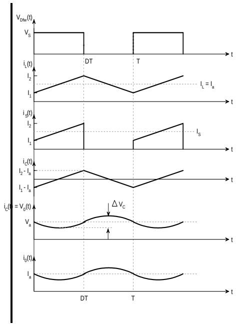
Continuous mode

- Efficiency

$$P_{loss} = P_{Q_s} + P_{D_{fw}} = I_o\left(\frac{t_{on}}{T}\right)V_{cesat} + I_o\left(1 - \frac{t_{on}}{T}\right)V_{D_{fw}}$$

$$\eta = \frac{P_{out}}{(P_{out} + P_{loss})} = \frac{V_a}{(V_a + D\,V_{cesat} + (1-D)V_{D_{fw}})} \; .$$

Buck converter
Critical inductance

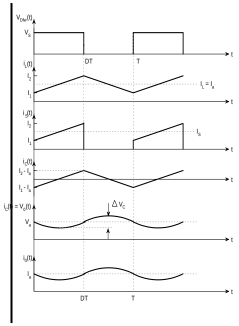


- If the load resistance is held constant and the inductance is reduced, the current ripple increases

- The value of L for which $i_L = 0$ at one and only one point per cycle is defined as the ***critical inductance*** L_c

Buck converter

Critical inductance



$$I_{L_p} = 2\,I_L = \Delta I = \frac{(V_s - V_a)t_{on}}{L_c} = \frac{(V_s - V_a)D}{f_s L_c} \;.$$

$$P_{in} = V_s\, I_s = V_s\, I_L\, D$$

$$P_{out} = \frac{V_a^2}{R} \;,$$

$$V_s\, I_L\, D = \frac{V_a^2}{R} \;.$$

$$I_L = \frac{\Delta I}{2} = \frac{(V_s - V_a)D}{2\, f_s L_c} \;.$$

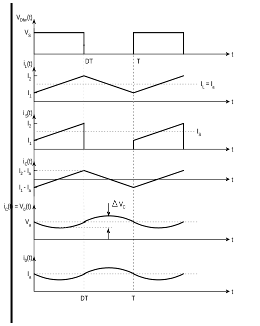
$$V_s \frac{(V_s - V_a)D^2}{2\, f_s L_c} = \frac{V_a^2}{R} \;.$$

$$\frac{V_s^2(1-D)D^2}{2\, f_s L_c} = \frac{V_a^2}{R} \;,$$

$$\frac{(1-D)D^2 V_s^2}{2\, f_s L_c} = \frac{V_s^2 D^2}{R} \;.$$

$$L_c = \frac{R(1-D)}{2\, f_s} \;.$$

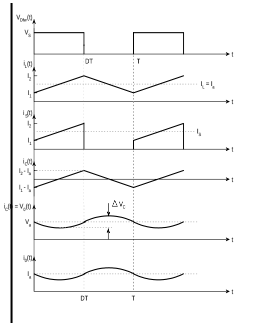
Buck converter
Critical load resistance



- If the inductance is held constant and the load resistance is increased, the inductor current decreases while keeping the ripple constant
- the buck converter will be operating in the discontinuous mode if the load resistance R is greater than its critical load resistance R_c

$$R_c = \frac{R_{nom}}{1 - D} \qquad R_{nom} = 2 f_s L$$

Buck converter
Discontinuous conduction mode

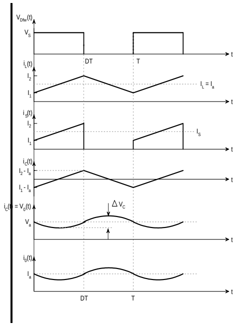


- If the load resistance is larger than the critical resistance

  or

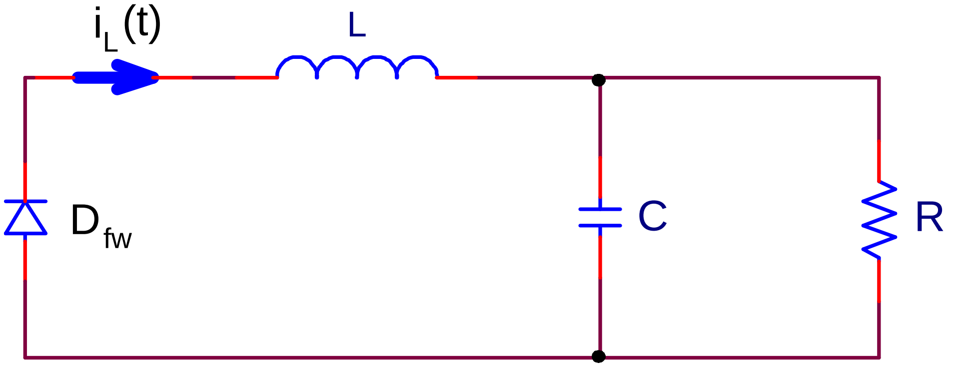
- If the inductance is smaller than the critical inductance

  the buck converter will operate in the discontinuous conduction mode

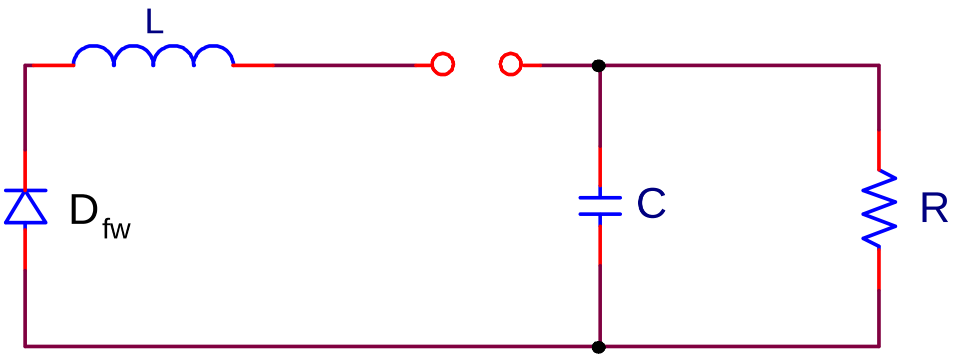


Buck converter

Discontinuous conduction mode

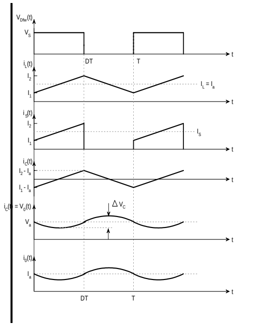


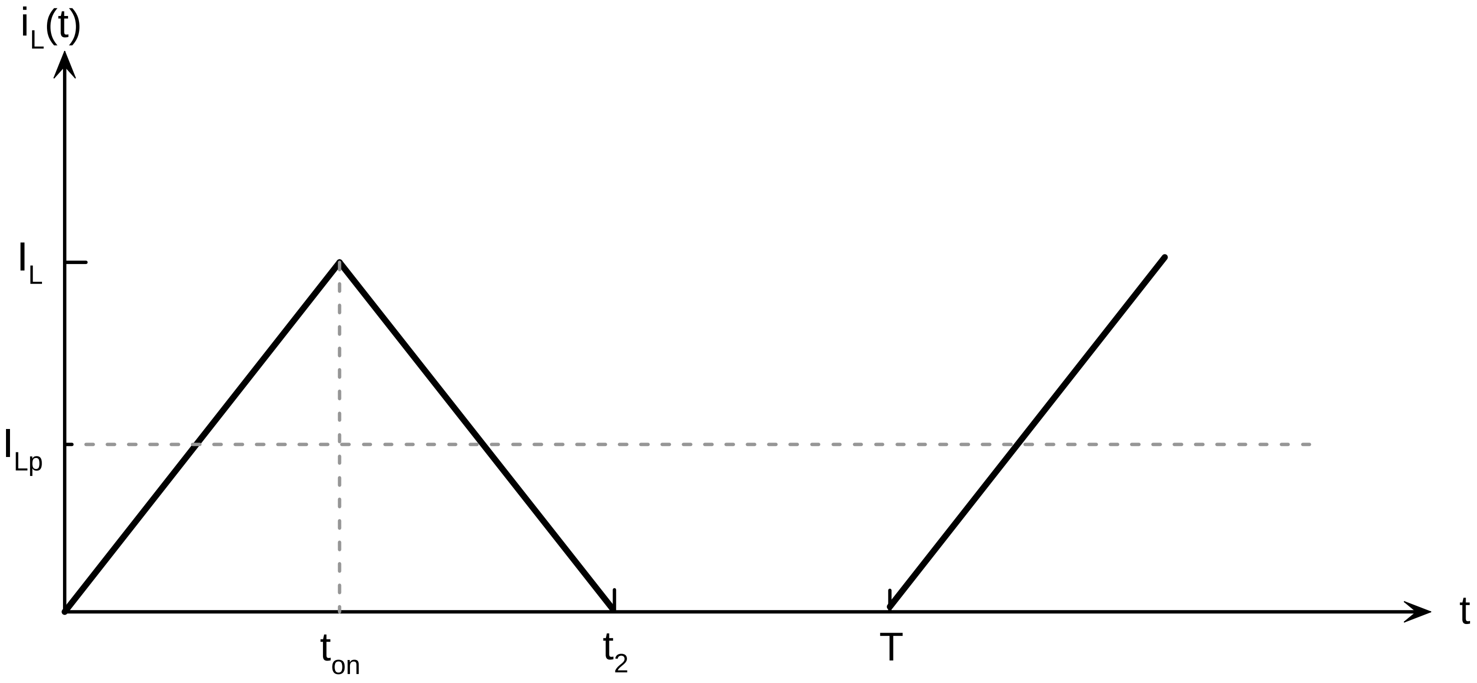
(a) $t_{on} < t \leq t_2$

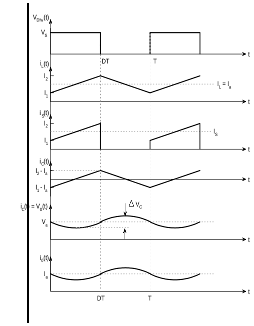


(a) $t_2 < t \leq T$

Buck converter
Discontinuous conduction mode







Buck converter
Discontinuous conduction mode

- Voltage conversion ratio

$$(V_s - V_a)\, t_{on}\, f_s - V_a\, (t_2 - t_{on})\, f_s = 0 \;.$$

$$\frac{I_{Lp}\, t_2\, f_s}{2} = \frac{V_a}{R} \;.$$

$$(V_s - V_a) = L\, \frac{I_{Lp}}{t_{on}} \;.$$

$$V_s\, t_{on}\, f_s - V_a\, t_2\, f_s = 0 \;.$$

Substituting $t_2 f_s$ and I_{Lp} into the above equation,

$$V_a^2 + V_a V_s \frac{D^2 L_c}{(1-D)L} - V_s^2 \frac{D^2 L_c}{(1-D)L} = 0 \;.$$

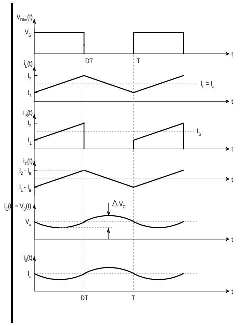
This is a quadratic equation

$$\frac{V_a}{V_s} = \frac{D\left[\sqrt{\frac{4L(1-D)}{L_c} + D^2} - D\right]}{\frac{2L(1-D)}{L_c}}$$

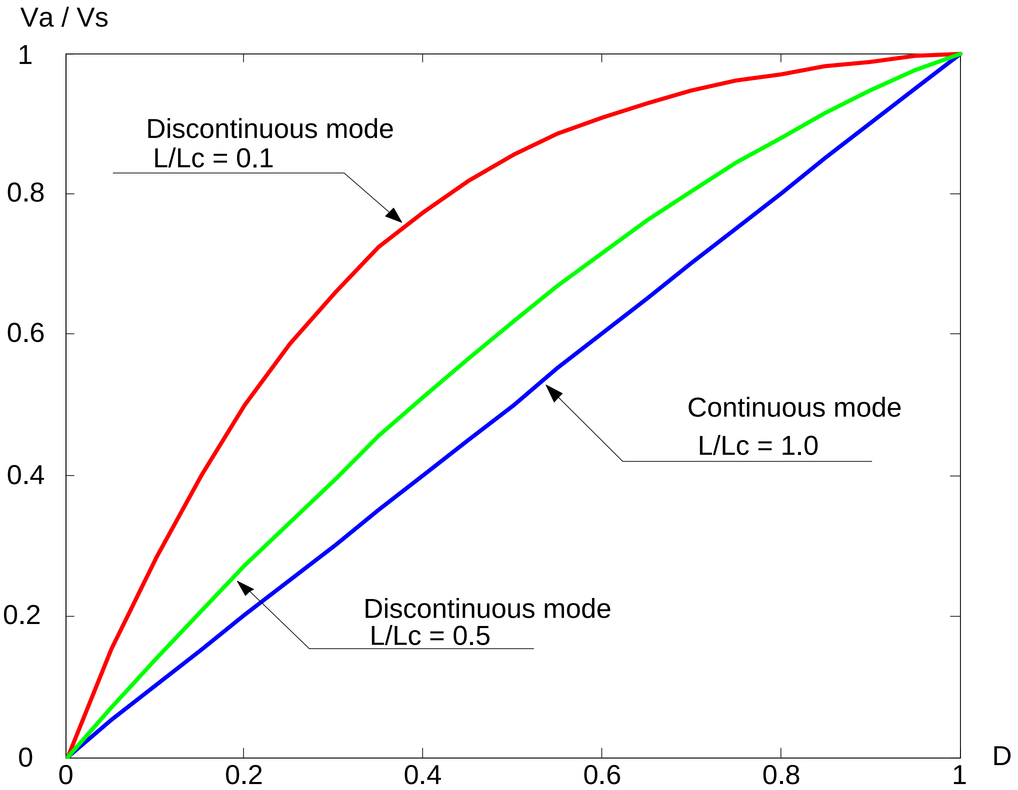
$$\frac{V_a}{V_s} = \frac{2}{1 + \sqrt{1 + \frac{4L(1-D)}{D^2 L_c}}} \;;\; L < L_c$$

Buck converter

Voltage conversion ratio

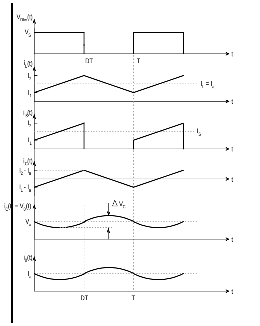


- Continuous mode
  - Linear
- Discontinuous mode
  - Non linear

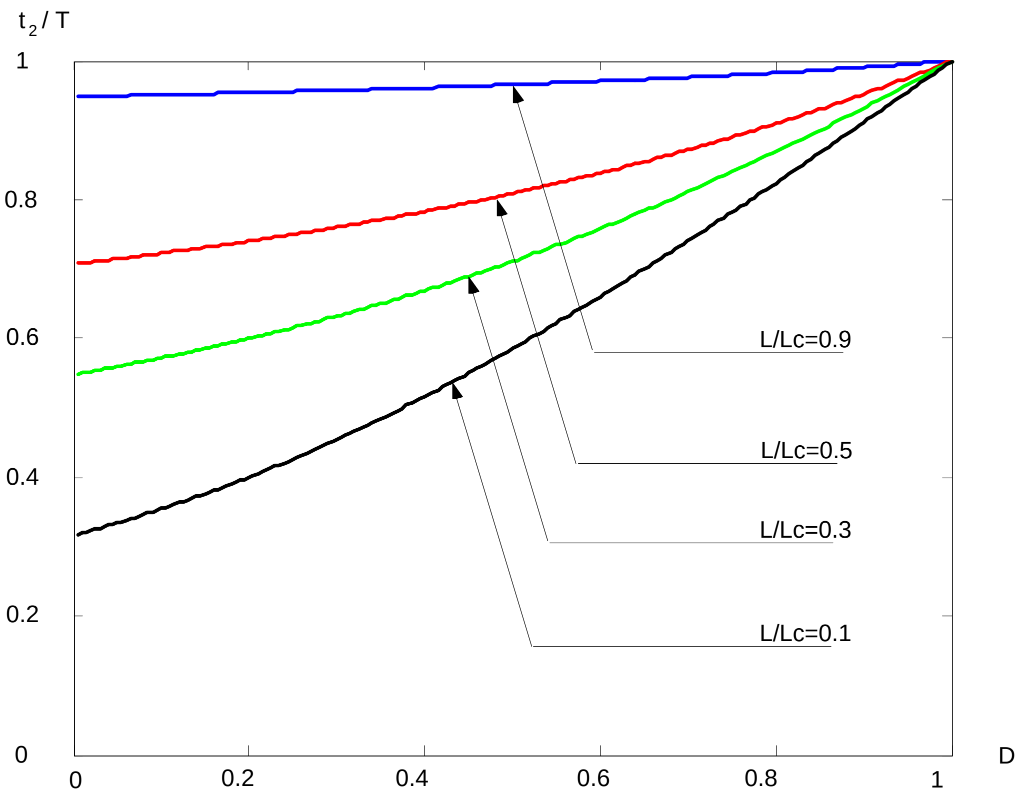


Buck converter

Discharge time in the discontinuous mode

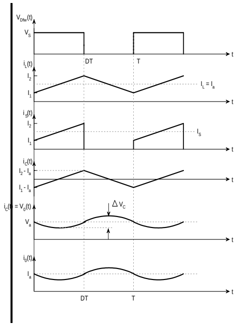


- Decreasing the inductance
  - Forces the converter deeper in the discontinuous mode
  - Decreases the discharge time
  - Reduces the losses on the diode
  - Increases the stresses

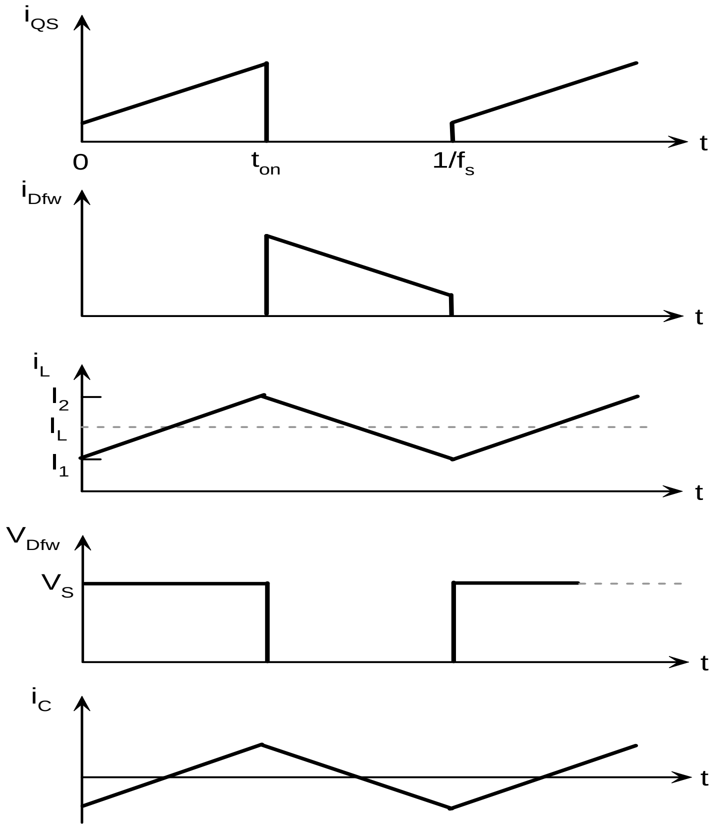


Buck converter

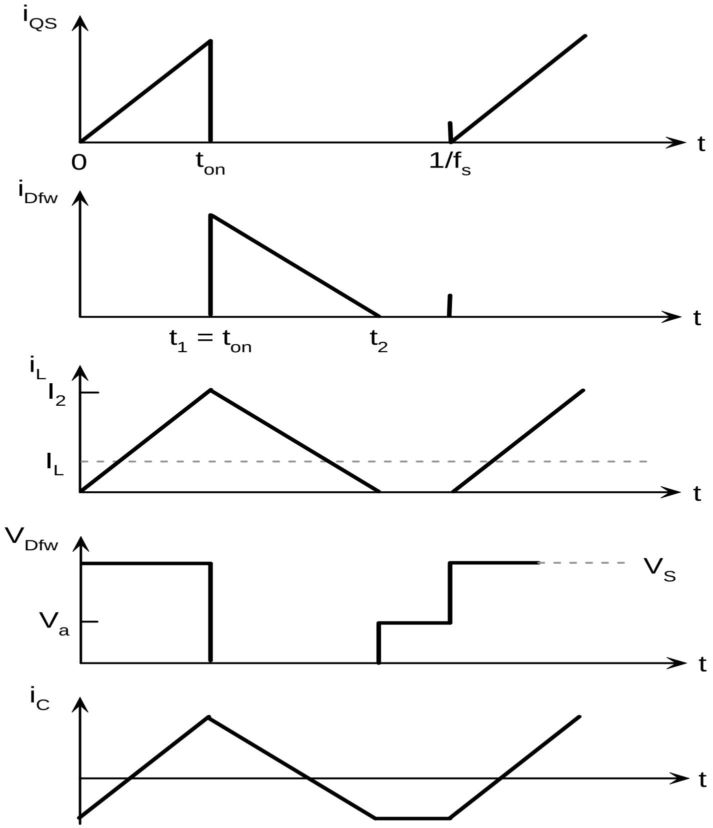
Switching waveforms



- Continuous mode
- Discontinuous mode

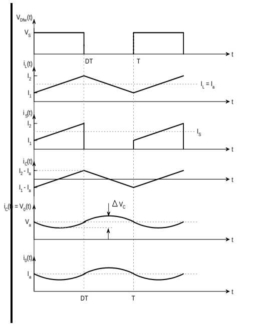


(a)



(b)

Buck converter



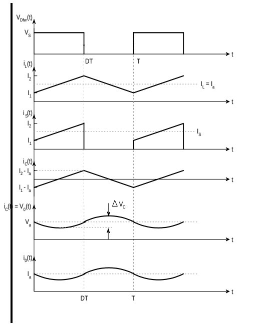
- Design the switching converter to operate either in the continuous or the discontinuous mode

- If the load of a buck converter is suddenly removed, the output voltage of an open-loop buck converter will rise to the same level as the input voltage

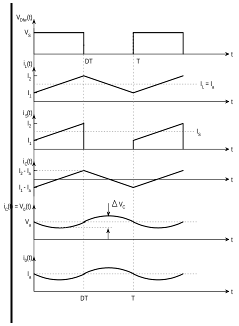
- The advantages of the buck converter are its ability to easily control output voltages and currents during turn-on and turn-off and under fault conditions

Buck converter

Example 2-1



- The buck converter with a bipolar switching transistor, as shown in Figure 2.1, has an input voltage of 10 V. The switching frequency is 1 kHz. The load requires an average voltage of 5 V with a maximum ripple voltage of 20 mV. The maximum ripple current of the output inductor is 0.2 A. Determine: (a) the duty cycle, (b) the output inductance, (c) the output capacitance, and (d) the output capacitance if the switching frequency is increased to 10 kHz.



Buck converter

Example 2-1

$$D = \frac{V_a}{V_s} = \frac{5}{10} = 0.5 \ .$$

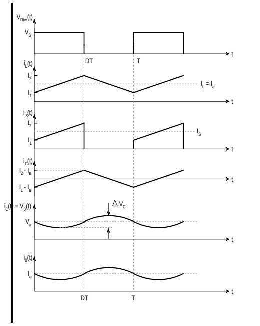
$$L = \frac{DV_s(1-D)}{f_s \Delta I} = \frac{0.5(10)(0.5)}{1000(0.2)} = 0.0125\ H = 12.5\ mH\ .$$

$$C = \frac{V_s D(1-D)}{8 f_s^2 L \Delta V_a} = \frac{10(0.5)(0.5)}{8(1000)^2 0.0125(0.02)} = 1250\ \mu F\ .$$

$$L = \frac{0.5(10)(0.5)}{10000(0.2)} = 0.00125\ H\ .$$

$$C = \frac{10(0.5)(0.5)}{8(10000)^2 0.00125(0.02)} = 125\ \mu F\ .$$

Synchronous Buck Converter

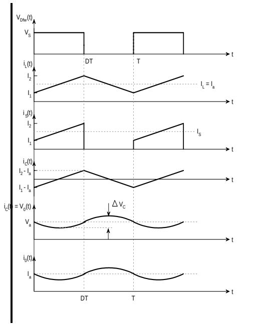


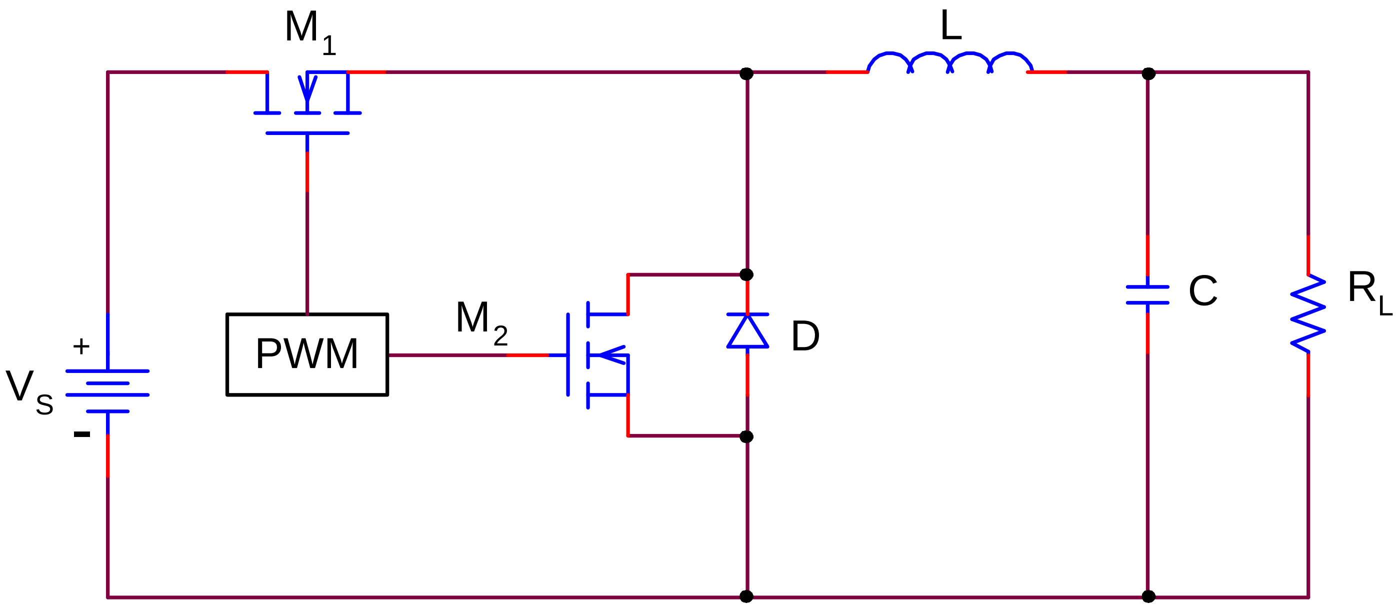
- When the output voltage of a buck converter is calculated, considering V_d yields:

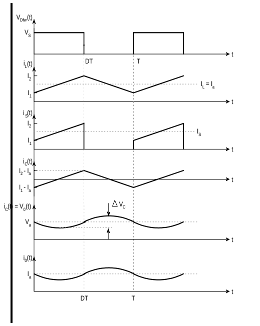
$$Va = DVs - (1 - D)Vd$$

- It is possible to replace the Schottky diode by a power MOSFET, this is the buck synchronous converter
- SBC works only in the continuous conduction mode

Synchronous Buck Converter

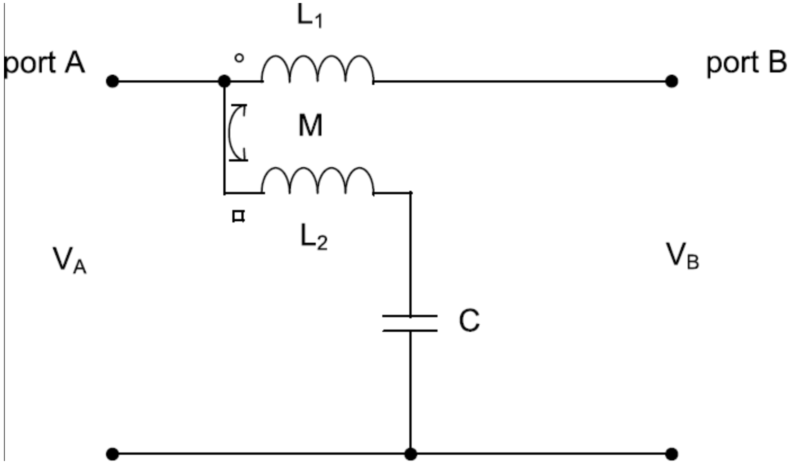




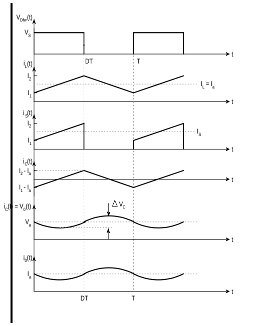


Ripple Steering

- Replace the series inductor with a pair of coupled inductors and a DC blocking capacitor to reduce the current ripple flowing through the inductor.
- Objective is to remove or filter the AC ripple present at port A to produce a clean DC voltage at port B by re-directing or steering the ripple current to an auxiliary branch of L_2 and C.

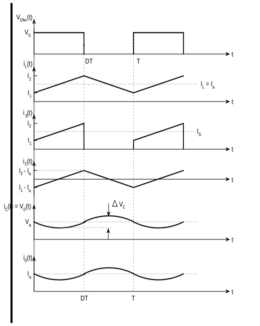


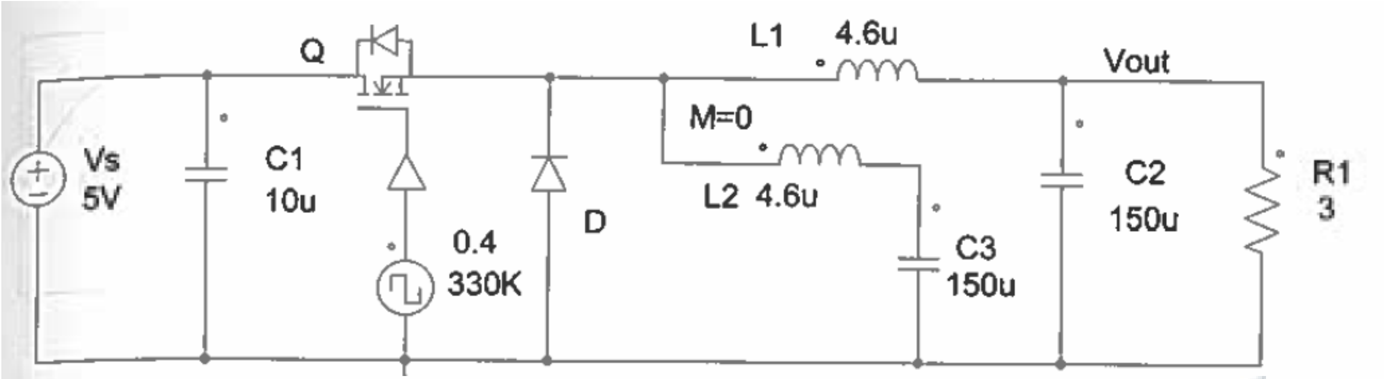
Ripple Steering

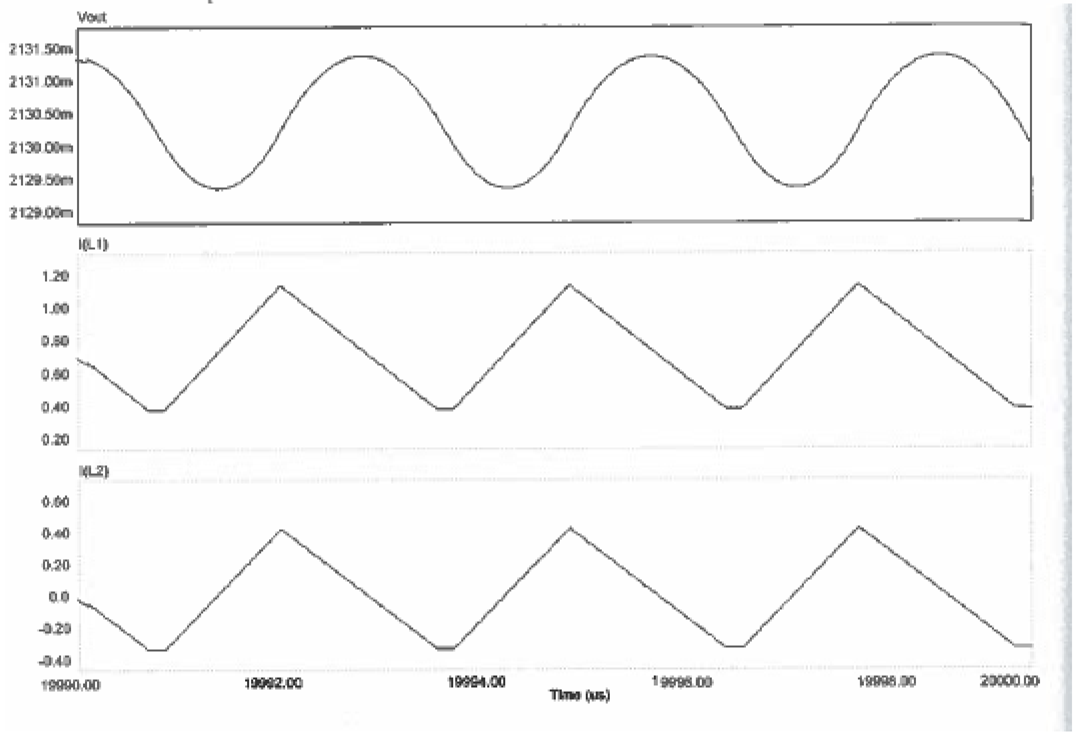


- Consider the case of a perfect coupling of L_1 and L_2 and $L_1 = L_2$. Also, the dc blocking capacitance C is infinity.
- Using KVL, we have $V_A = V_{Aac} + V_{Adc}$
- Since the capacitance of C is infinity, its AC impedance is zero. Therefore, $V_C = V_{Adc}$.
- Thus, the voltage drop across L_2 is $V_{L2} = V_{Aac}$.
- For a perfect coupling of L_1 and L_2, $V_{L1} = V_{L2} = V_{Aac}$.
- Hence, the out port B voltage is $V_B = V_{Adc}$.
- The output port B is a pure DC voltage since the ripple current has been steered to the auxiliary branch of L_2 and C.

Buck Converter with LC Branch







Buck Converter with Coupled L